IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected copy

DIGITAL SIGNAL PROCESSING

Monday, 14 December 9:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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DIGITAL SIGNAL PROCESSING

1. a) Given a discrete-time signal $x(n)$, write down the formula for the z-transform $X(z)$ and hence find the z-transform of

$$x(n) = [-10,\ \underset{\uparrow}{10},\ 5,\ 2].$$

. [2]

b) Consider a linear system with system function $H(z)$. Explain what is meant by the term *Region of Convergence* in the context of the z-transform and state the relationship between the *Region of Convergence* and the *stability* of $H(z)$. [2]

c) Next consider

$$P(z) = \frac{1}{1 - pz^{-1}}$$

and the unit step function $u(n)$.

i) If the inverse z-transform of $P(z)$ corresponds to a causal signal, write an expression in the discrete-time domain for this causal signal and state the *Region of Convergence.* [1]

ii) If the inverse z-transform of $P(z)$ corresponds to an anticausal signal, write an expression in the discrete-time domain for this anticausal signal and state the *Region of Convergence.* [1]

d) For the causal system

$$Q(z) = \frac{5z + 15.4}{z^2 + 5.2z + 1}$$

find the inverse z-transform of $Q(z)$ and explain whether or not $Q(z)$ is stable. [5]

e) The block diagram of Figure 1.1 shows a discrete-time system with system function $H(z)$. The z-transforms of the input signal $x(n)$ and the output signal $y(n)$ are $X(z)$ and $Y(z)$ respectively. Evaluate the system function $H(z)$ in terms of the constant scalar coefficients d_1, d_2 and state the key property of $H(z)$. [9]

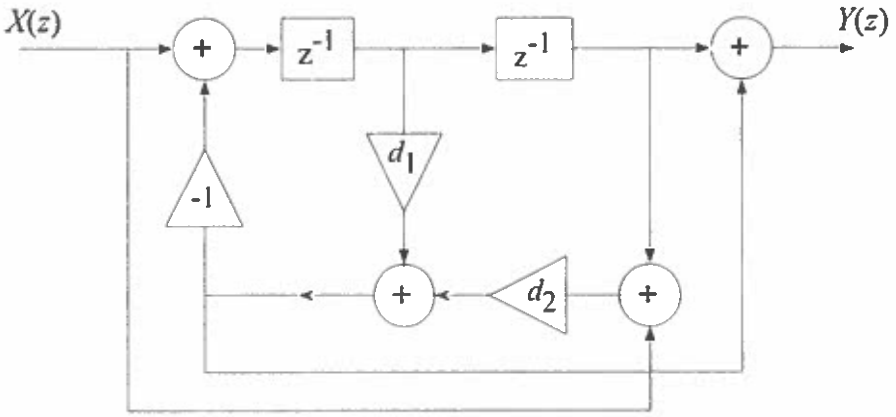


Figure 1.1 Signal flow graph

2. Consider a discrete-time filter with finite impulse response for which the input is denoted $x(n)$, the output is denoted $y(n)$, n is the discrete-time index and

$$y(n) = -0.5x(n) - 0.45x(n-2).$$

a) Draw a labelled sketch plot of the z-plane and indicate on the plot the positions of the poles and zeros of this filter. [4]

b) Write an expression for the transfer function of the filter. [2]

c) Write an expression for the magnitude of the frequency response of this filter. [3]

d) Draw a labelled sketch of the magnitude of the frequency response of this filter and mark on the sketch the values of the magnitude in dB at frequencies of 0, $\pi/2$, and π. [6]

e) Define group delay for discrete-time filters and estimate the group delay of this filter in seconds at a frequency of 2 kHz given that the sampling frequency is 16 kHz. [5]

3. a) Consider the discrete-time signal $x(n) = 2a^n u(n)$ where $u(n)$ is the unit step function and $|a| < 1$.

i) Write down an expression for the spectrum of this signal. [2]

ii) In an example of a multirate signal processing system, the signal $x(n)$ is decimated by a factor of 2. Explain, with an illustrative sketch, the effect that such decimation has in the frequency domain and hence determine the spectrum of the signal after downsampling. [3]

b) i) State and explain the Noble Identities. [3]

ii) Figure 3.1 shows two multirate signal processing systems.

Denoting the input samples as $x(n) = \{x0, x1, x2, \ldots\}$ and $y(n) = \{y0, y1, y2, \ldots\}$, find the corresponding samples at the points in the Figure marked A, C and, hence, the first 6 samples at each of the points marked B and D. Comment on the importance of the order of operation of multirate processing blocks such as those in Figure 3.1. [6]

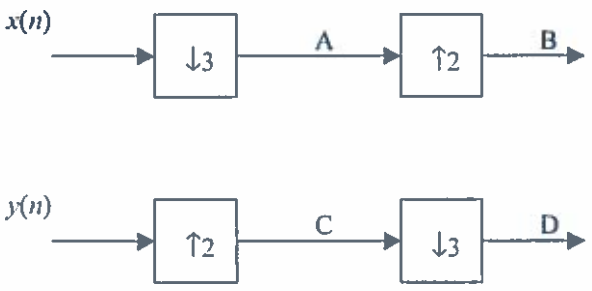


Figure 3.1 Multirate signal processing systems.

c) Consider a DSP system operating on an input signal $x(n)$ sampled at a rate of 1000 samples per second. Using multirate signal processing techniques, design a system to delay $x(n)$ by 300 μs. Show your design in terms of a labelled block diagram together with a detailed explanation of its operation. [6]

4. Consider a discrete-time signal $x(n)$ of length N samples and having N-point DFT $X(k)$.

a) If $X(k)$ is a real sequence, what conditions must be satisfied by $x(n)$, assuming $x(n)$ is real? Give an example for $x(n)$ of length 6 samples for which $X(k)$ is real. [4]

b) Let $x(n) = [1, -1, 0, 2]$. Calculate $X(k)$. [6]

c) If $x(n)$ satisfies the condition $x(n) = x(N-1-n)$ and N is an even number, show that $X(N/2) = 0$. [3]

d) If $x(n) = -x(N-1-n)$, show that $X(0) = 0$. [3]

e) Now consider the magnitude and phase of $X(k)$. If $x(n)$ is real, what conditions must be satisfied by the magnitude and phase of $X(k)$. Draw a labelled sketch of the magnitude and phase spectra of $X(k)$ for an illustrative example of a case for which $x(n)$ is real. [4]